

AKM Iqtidar Newaz

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🎓 Google Scholar 🌐 Personal Website

Lecturer, North South University

Research Interests

- Web Security and Privacy
- Browser Extension Security and Privacy
- Security of Internet of Things (IoT)

Education

- 2021 **Master of Computer Science**, *Florida International University, USA*
2014 **Bachelor of Computer Science**, *Islamic University of Technology, Bangladesh*

Grants & Awards

- 2025 Soteria: NSU Cybersecurity Center for Enhancing Cybersecurity Skills among Students and Local Communities to Help Safeguard Them against Online Threats (\$100,000 as Co-PI)
- 2024–2025 Conference Travel & Research Grant (CTRG) for *Mnemosyne* project where I am exploring the idea of fingerprinting browser extensions through global memory location writes
- 2022 Recipient of the RSA Conference Security Scholar
- 2019 Student travel grant from Graduate & Professional Student Committee (GPSC) for IEEE SNAMS
- 2018 Recipient of the Sweden-Bangladesh Trust Fund Travel Grant
- 2011–2014 Full free OIC scholarship for undergraduate study in Islamic University of Technology
- 2008 Government scholarship from the Board of Secondary Education, Dhaka (Merit Position 52)

Publications

Md. Rayhan Kabir Khan, Md Fahim Shahoriar Titu, Istiak Ahammed, **AKM Iqtidar Newaz**, and Riasat Khan, "Janus: A Novel Secret Key Generation Framework Combining W-State QKD and Post-Quantum Authentication" 23rd International Conference on Security and Cryptography (SECRYPT), 2026

AKM Iqtidar Newaz, Ahmet Aris, Amit Kumar Sikder, and A. Selcuk Uluagac, "A Systematic Threat Analysis of Modern Healthcare Communication Systems" IEEE Global Communications Conference (GLOBECOM), 2022

AKM Iqtidar Newaz, Amit Kumar Sikder, Mohammad Ashiqur Rahman, and A. Selcuk Uluagac, "A Survey on Security and Privacy Issues in Modern Healthcare Systems: Attacks and Defenses" ACM Transactions on Computing for Healthcare (HEALTH), 2021

AKM Iqtidar Newaz, Nur Imtiazul Haque, Amit Kumar Sikder, Mohammad Ashiqur Rahman, and A. Selcuk Uluagac, "Adversarial Attacks to Machine Learning-Based Smart Healthcare Systems" IEEE Global Communications Conference (GLOBECOM), 2020

AKM Iqtidar Newaz, Amit Kumar Sikder, Leonardo Babun, and A. Selcuk Uluagac, "HEKA: A Novel Intrusion Detection System for Attacks to Personal Medical Devices" IEEE Conference on Communications and Network Security (CNS), 2020

AKM Iqtidar Newaz, Amit Kumar Sikder, and A. Selcuk Uluagac, "Healthguard: A machine learning-based security framework for smart healthcare systems" IEEE Sixth International Conference on Social Networks Analysis, Management and Security, 2019

Adnan Ferdous Ashrafi, **AKM Iqtidar Newaz**, Rasif Ajwad, Moain Mahmud Tanvee, and M. A. Mottalib, "A modified algorithm for DNA motif finding considering variable length motif and mutation" IEEE 2nd International Conference on Recent Trends in Information Systems (ReTIS), 2015

Professional Experience

North South University (Lecturer, ECE Department)

- 2023–
- Instructor for Data Structure & Algorithms (CSE 225).
 - Instructor for Design & Analysis of Algorithms (CSE 311).
 - Instructor for Software Engineering (CSE 327).
 - Instructor for Introduction to Cybersecurity (CSE 146).

North South University CyberSecurity Center (Executive Director)

- 2026–
- Developed a 12 hours course contents for the undergraduate students and MSMEs.
 - Instructor for Digital Hygiene and Cybersecurity Awareness course.

Florida International University (Graduate Research & Teaching Assistant)

- 2017–2022
- Security analysis of Healthcare IoT devices.
 - Security analysis of Healthcare Communication Protocols.
 - Teaching assistant for Network Security (EEL 6877).
 - Teaching assistant for Introduction to Security of Internet of Things (EEL 4717).

Robi Axiata Limited (Specialist, Service Charging Operation Department)

- 2015–2017
- Managed online and batch service provisioning functions to ensure business objectives.
 - 3G network monitoring and troubleshooting and design test projects for LTE implementation.

Other Activities

Training & Certifications

- 2025 Training of Trainers on "How To Make General Employees Aware of Cybersecurity", JICA, BASIS, BCC, ICT DIVISION.
- 2024 Training of Trainers on "B-TopSE Program Architecture Course", JICA, BASIS, BCC, ICT DIVISION.

Poster Presentation

- 2020 AKM Iqtidar Newaz, Ahmet Aris, and A. Selcuk Uluagac, "Adversarial Attacks to Machine Learning-Based Smart Healthcare Systems" 5th Annual Big Data Conference: Big Data, Business, and the Pandemic, University of Miami.

Organizer Role

- 2020 Session chair in IEEE Conference on Communications and Network Security (CNS).
- 2019 Student volunteer in 12th ACM Conference on Security and Privacy in Wireless and Mobile Networks.
- 2018 Student volunteer in IEEE Intelligence and Security Informatics (ISI).

Reviewer Role

- 2025 IEEE Transactions on Information Forensics and Security
- 2020–2022 IEEE Communication Surveys and Tutorials (IEEE COMST)
- 2020–2022 IEEE Transactions on Mobile Computing (TMC)
- 2022 IEEE Access
- 2020–2022 IEEE Internet of Things
- 2023 Wireless networks (WN); the journal of mobile communication, computation and information
- 2021–2022 IEEE Global Communications Conference (GLOBECOM)
- 2021 IEEE European Symposium on Security and Privacy (Euro S&P)
- 2021 IEEE Conference on Communications and Network Security (CNS)

Skills

- Operating Systems Linux Ubuntu, Windows, Raspbian, Android
- Languages Python, Java, C/C++, Assembly language, SQL
- Security Tools Wireshark, Scapy, TCPDump, OpenSSH/PuTTY, IDA Pro, Ollydbg

References

Available upon request.